MAT 4233 — Weekly Schedule

Modern Abstract Algebra · Fall 2026 · UT San Antonio

Texts

Primary text

- **LiZh** — Li & Zhao, *Introduction to Abstract Algebra*

Secondary reading

- **KvP** — Kramer & von Pippich, *From Natural Numbers to Quaternions*

Handouts & papers

- **Conrad** — Conrad, *The Gaussian Integers*

Schedule

Date	Topic	Reading
Thu 20 Aug	Lecture — Course launch. The integers: division algorithm, gcd, Bezout; congruence and the ring Zn	KvP §1.1-1.5
Tue 25 Aug	Lecture — Binary operations; Cayley tables; Zn under addition and multiplication	LiZh §1.1
Thu 27 Aug	Lecture — Isomorphic binary structures	LiZh §1.2
Tue 1 Sep	Lecture — Groups: axioms and first properties. Zn, the unit group (Zn)*, and the symmetries of a square	LiZh §1.3
Thu 3 Sep	Lecture — Subgroups; subgroup criteria; the subgroups of Dn and of Zn	LiZh §1.4
Tue 8 Sep	**Q&A** (**Due**: Set 1)	
Thu 10 Sep	Lecture — Cyclic groups; the order of an element; Zn, (Zp)* and primitive roots	LiZh §1.5
Tue 15 Sep	Lecture — Generating sets; Dn as the group generated by a rotation and a reflection	LiZh §1.6
Thu 17 Sep	Lecture — Permutation groups; cycle notation; Cayley's theorem	LiZh §2.1
Tue 22 Sep	Lecture — Alternating groups; the sign of a permutation; A4 and the tetrahedron	LiZh §2.2
Thu 24 Sep	Lecture — Plane isometries and finite symmetry groups: every one is cyclic or dihedral	Instructor notes
Tue 29 Sep	**Q&A** (**Due**: Set 2)	
Thu 1 Oct	**Exam** — Midterm Exam 1 — Sets 1 and 2	
Tue 6 Oct	Lecture — Cosets; Lagrange's theorem; index	LiZh §3.1
Thu 8 Oct	Lecture — Homomorphisms: kernel, image, and first properties	LiZh §3.3
Thu 15 Oct	Lecture — Normal subgroups; quotient groups; Z/nZ built properly	LiZh §3.4
Tue 20 Oct	Lecture — The isomorphism theorems; groups of small order	LiZh §3.4

Date	Topic	Reading
Thu 22 Oct	Lecture — Rings and subrings; Zn as a ring; units and zero divisors	LiZh §4.1
Tue 27 Oct	**Q&A** (**Due**: Set 3)	
Thu 29 Oct	Lecture — Integral domains and fields; Zn is a field exactly when n is prime; characteristic. Euler's and Fermat's theorems via (Zn)*	LiZh §4.2; LiZh §4.6
Tue 3 Nov	Lecture — Ring homomorphisms, ideals and quotient rings; the isomorphism theorems for rings	LiZh §4.5
Thu 5 Nov	Lecture — Ideal theory: principal, prime and maximal ideals	LiZh §4.7
Tue 10 Nov	Lecture — Irreducibles and primes; unique factorization domains; principal ideal domains	LiZh §5.1-5.2
Thu 12 Nov	Lecture — Euclidean domains; the Euclidean algorithm in Z and in k[x]	LiZh §5.3
Tue 17 Nov	Lecture — Multiplicative norms; the Gaussian integers Z[i]; Fermat's Two Squares theorem	LiZh §5.5; Conrad (statements and examples; proofs optional)
Thu 19 Nov	**Q&A** (**Due**: Set 4)	
Tue 24 Nov	Lecture — Sums of three and four squares; Z[sqrt(-5)] and the failure of unique factorization — a closing survey	Instructor notes; KvP §5.1-5.2
Tue 1 Dec	*Review* — Review — the exam pool	
Thu 3 Dec	**Exam** — Midterm Exam 2 — Sets 3 and 4	